

Math Foundations: Information Theory

Paired reading: 02_readings/optimization_uncertainty/ (3 PDFs) **Focus area A** (flagged weak area)

Information theory is unusual: it is both a **Week-0 math prerequisite** and a full lecture (**OL3**). This file is the compact refresher / formula sheet plus the derivations and worked examples you need so SL2's decision trees do not ambush you before OL3 arrives. For the No Free Lunch theorem and the full conceptual treatment, see ../02_optimization_uncertainty/OL3_information_theory.md.

1 The Intuition: Information as Surprise

Define the **surprisal**, or self-information, of an outcome x with probability $p(x)$:

$$I(x) = -\log_2 p(x) \quad \text{bits.}$$

Why this functional form? Shannon's three desiderata for any measure of "information":

1. **Continuous** in p .
2. **Monotonic:** rarer events should be more surprising: $p(x) \downarrow \Rightarrow I(x) \uparrow$.
3. **Additive for independent events:** $I(x, y) = I(x) + I(y)$ whenever $p(x, y) = p(x)p(y)$.

The third forces a logarithm: the only function with $f(pq) = f(p) + f(q)$ is $f(p) = c \log p$, and the sign plus base just fix units: $\log_2$ gives bits, while $\ln$ gives nats.

A few sanity points along the curve:

- Certain event ($p = 1$): $I = 0$, no surprise.
- Fair coin ($p = 1/2$): $I = 1$ bit per outcome.
- One-in-a-million event: $I \approx 20$ bits, which is why unusual events feel informationally heavy.

2 The Four Quantities: Definitions and Derivations

2.1 Entropy $H(X)$

$$H(X) = \mathbb{E}[I(X)] = -\sum_x p(x) \log_2 p(x),$$

with differential entropy using an integral in the continuous case.

Read it as **expected surprise**, or equivalently the average number of bits needed to encode X , by Shannon's source coding theorem.

Bounds for discrete X with n outcomes:

$$0 \leq H(X) \leq \log_2 n.$$

- $H = 0$ when X is deterministic, so one outcome has $p = 1$.
- $H = \log_2 n$ when X is **uniform**: most uncertain, hardest to predict, longest expected code.

Why uniform maximizes entropy: one-line Lagrangian. Maximize $-\sum_i p_i \log p_i$ subject to $\sum_i p_i = 1$. Lagrangian:

$$L = -\sum_i p_i \log p_i - \lambda \left(\sum_i p_i - 1 \right),$$

$$\frac{\partial L}{\partial p_i} = -\log p_i - 1 - \lambda = 0 \quad \Rightarrow \quad p_i = e^{-1-\lambda}.$$

All p_i are equal, so the solution is uniform. This is the **principle of maximum entropy**: in absence of constraints, the least-committal distribution is uniform.

2.2 Joint and Conditional Entropy

$$H(X, Y) = -\sum_{x,y} p(x, y) \log p(x, y),$$

$$H(Y | X) = \sum_x p(x) H(Y | X = x) = -\sum_{x,y} p(x, y) \log p(y | x).$$

The **chain rule of entropy**:

$$H(X, Y) = H(X) + H(Y | X) = H(Y) + H(X | Y).$$

This follows directly from $\log p(x, y) = \log p(x) + \log p(y | x)$ inside the expectation. Key inequality:

$$H(Y | X) \leq H(Y) \quad (\text{conditioning never increases entropy on average}).$$

Equality holds iff $X \perp Y$. Intuition: on average, knowing something can only reduce your uncertainty, never increase it. For a specific value $X = x$, $H(Y | X = x)$ can exceed $H(Y)$; the average over x cannot.

2.3 Mutual Information $I(X; Y)$

$$I(X; Y) = H(Y) - H(Y | X) = H(X) - H(X | Y),$$

$$I(X; Y) = H(X) + H(Y) - H(X, Y),$$

$$I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}.$$

All four expressions are equal, and each is useful depending on what you have.

Properties:

$$I(X; Y) \geq 0, \quad I(X; Y) = 0 \iff X \perp Y,$$

$$I(X; Y) = I(Y; X), \quad I(X; X) = H(X).$$

Mutual information is the **information shared** between X and Y : how much knowing one reduces your uncertainty about the other.

2.4 KL Divergence $D_{\text{KL}}(p\|q)$

$$D_{\text{KL}}(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = \mathbb{E}_p[\log p(X) - \log q(X)].$$

This is **the number of extra bits you pay** if you encode samples from p using an optimal code designed for q .

Properties:

$$D_{\text{KL}}(p\|q) \geq 0, \quad D_{\text{KL}}(p\|q) = 0 \iff p = q \text{ almost everywhere,}$$

$$D_{\text{KL}}(p\|q) \neq D_{\text{KL}}(q\|p).$$

KL is asymmetric and is not a distance metric.

Proof that KL is nonnegative via Jensen's inequality. $\log$ is concave, so for any random variable Z , $\mathbb{E}[\log Z] \leq \log \mathbb{E}[Z]$. Let $Z = q(X)/p(X)$ under p :

$$\begin{aligned} -D_{\text{KL}}(p\|q) &= \mathbb{E}_p \left[\log \frac{q(X)}{p(X)} \right] \\ &\leq \log \mathbb{E}_p \left[\frac{q(X)}{p(X)} \right] \\ &= \log \sum_x p(x) \frac{q(x)}{p(x)} \\ &= \log \sum_x q(x) \\ &= \log 1 = 0. \end{aligned}$$

So $-D_{\text{KL}} \leq 0$, i.e. $D_{\text{KL}} \geq 0$. Equality holds iff q/p is constant p -almost-everywhere, i.e. $p = q$.

2.5 Cross-Entropy

$$H(p, q) = - \sum_x p(x) \log q(x) = H(p) + D_{\text{KL}}(p\|q).$$

Reading: the average number of bits to encode samples from p using a code optimized for q . Two pieces: $H(p)$, the irreducible entropy of the generating distribution, and $D_{\text{KL}}(p\|q)$, the inefficiency due to using the wrong code.

Consequence: minimizing cross-entropy over q is identical to minimizing $D_{\text{KL}}(p\|q)$, because $H(p)$ does not depend on q . So neural-network classification, which minimizes cross-entropy loss, is **literally** finding the model q whose distribution is closest in KL to the empirical $\hat{p}_{\text{data}}$.

3 The Information-Theoretic Identity Sheet

$H(X, Y)$	$= H(X) + H(Y X)$	chain rule
$H(Y X)$	$\leq H(Y)$	conditioning reduces entropy on average
$I(X; Y)$	$= D_{\text{KL}}(p(x, y) \ p(x)p(y))$	MI as KL
$I(X; Y)$	$= H(Y) - H(Y X)$	information gain form
$H(p, q)$	$= H(p) + D_{\text{KL}}(p\ q)$	cross-entropy decomposition.

If a question asks “rewrite this in a different form,” these are the substitutions you reach for.

4 Information Gain: The Bridge to SL2's Decision Trees

Information gain is mutual information by another name. For a decision tree splitting a labeled set S on attribute A :

$$\text{Gain}(S, A) = H(S) - H(S | A) = H(S) - \sum_v \frac{|S_v|}{|S|} H(S_v) = I(\text{label}; \text{attribute}).$$

Where $H(S) = -\sum_c p_c \log p_c$ is the entropy of the label distribution in S , and S_v is the subset where $A = v$. ID3 picks the attribute maximizing this: the attribute whose values are most informative about the label.

4.1 Worked Example

12 emails: 8 spam, 4 ham. Feature contains “free”: of those, 6 spam / 1 ham say “yes” (7 total), and 2 spam / 3 ham say “no” (5 total).

Parent entropy with proportions 8/12 spam and 4/12 ham:

$$p = \frac{8}{12} \approx 0.667,$$

$$H(S) = -0.667 \log_2 0.667 - 0.333 \log_2 0.333 \approx 0.390 + 0.528 = 0.918 \text{ bits.}$$

Child entropies:

$$\text{“yes” subset (6 spam, 1 ham, } n = 7\text{): } p = \frac{6}{7},$$

$$H = -\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} \approx 0.592 \text{ bits.}$$

$$\text{“no” subset (2 spam, 3 ham, } n = 5\text{): } p = \frac{2}{5},$$

$$H = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} \approx 0.971 \text{ bits.}$$

Weighted child entropy:

$$H(S | A) = \frac{7}{12} \cdot 0.592 + \frac{5}{12} \cdot 0.971 \approx 0.345 + 0.405 = 0.750 \text{ bits.}$$

Gain:

$$\text{Gain} = 0.918 - 0.750 = 0.168 \text{ bits.}$$

That number is exactly $I(\text{label}; \text{“free”})$. The split is informative: knowing whether the email contains “free” reduces label uncertainty by 0.168 bits. ID3 compares this number across candidate attributes and picks the largest.

4.2 Why Entropy as the Impurity Measure?

ID3's choice of entropy is not arbitrary. The split criterion needs to be zero at pure nodes, maximal at maximally mixed nodes, and concave in the class probabilities. Three common choices satisfy this:

Measure	Formula	Used by
Entropy	$-\sum_i p_i \log p_i$	ID3, C4.5
Gini impurity	$1 - \sum_i p_i^2$	CART
Misclassification error	$1 - \max_i p_i$	rarely; not strictly concave

Entropy and Gini almost always pick the same split; they are locally proportional near uniform.

5 Maximum Entropy Principle: The OL3 Idea

When choosing a distribution under constraints, **pick the one with maximum entropy** subject to those constraints. The justification is operational: this is the distribution that commits to the least beyond what has been specified.

Worked cases:

Constraint(s)	Max-entropy distribution
$\sum p = 1$ only	Uniform over outcomes
Mean μ , support $[0, \infty)$	Exponential ($1/\mu$)
Mean μ , variance σ^2 , support $\mathbb{R}$	Gaussian $\mathcal{N}(\mu, \sigma^2)$
Mean μ , support $\{0, 1, \dots\}$	Geometric

This is the principled reason the Gaussian is the “uninformative” continuous model with known mean and variance, and why “uniform prior” is the default for “no prior information.”

6 KL Asymmetry: Why Direction Matters

$D_{\text{KL}}(p\|q)$ and $D_{\text{KL}}(q\|p)$ are different, and the choice changes optimization behavior:

- $D_{\text{KL}}(p\|q)$ (**forward KL**): penalizes q heavily wherever p has mass but q does not, since $p \log(p/q) \rightarrow \infty$ as $q \rightarrow 0$. Result: q becomes **mean-seeking**, broad and mass-covering.
- $D_{\text{KL}}(q\|p)$ (**reverse KL**): penalizes q wherever it has mass but p does not. Result: q becomes **mode-seeking**, narrow and focused on one mode.

Where this comes up:

- **Cross-entropy classification loss** uses forward KL: the model must put mass everywhere the data does.
- **Variational inference** uses reverse KL: q is fit to a single mode of the true posterior.
- **t-SNE** uses an asymmetric divergence that is effectively reverse-ish: it preserves local structure (modes) at the expense of global geometry, which is exactly why t-SNE plots look the way they do.

7 The No Free Lunch Theorem: OL3 Brief Pointer

Wolpert and Macready, 1997: averaged over **all** possible target functions, no optimization or learning algorithm performs better than any other. Every algorithm trades favorable performance on some problems for unfavorable performance on others.

Practical reading: there is no universally best ML algorithm. Your prior or bias about the problem class is what makes one work. This is why SL1’s “bias is a feature, not a bug” lecture exists. Full development lives in `../02_optimization_uncertainty/OL3_information_theory.md`.

8 Worked Sanity Checks

Distribution	H (bits)
Fair coin ($p = 0.5$)	$-0.5(-1) - 0.5(-1) = 1.000$
Biased coin ($p = 0.9$)	$-0.9 \log_2 0.9 - 0.1 \log_2 0.1 \approx 0.469$
Biased coin ($p = 0.25$)	$-0.25 \log_2 0.25 - 0.75 \log_2 0.75 \approx 0.811$
Certain event ($p = 1$)	0
Fair 8-sided die	$\log_2 8 = 3.000$
Fair 6-sided die	$\log_2 6 \approx 2.585$

If you can reproduce these four in your head, the SL2 entropy/gain computations will not slow you down.

9 Where Each Quantity Resurfaces

Quantity	Appears in
Entropy / information gain	SL2 decision-tree splits; UL2 filter feature selection
Mutual information	OL1 MIMIC's dependency tree; UL2 filters; UL3 ICA objective
Conditional entropy	SL2 weighted-child impurity
KL divergence	OL4 (MLE = minimize KL to data); variational bounds; UL4 t-SNE
Cross-entropy	Standard NN classification loss (SL3); logistic regression
Max-entropy = uniform	OL3 max-entropy principle; OL4 uninformative prior
Max-entropy = Gaussian	OL4 default noise model

10 Review Questions with Sketched Answers

Conceptual Checks

1. One-sentence definitions.

Entropy: expected surprise / minimum bits to encode a source. **Conditional entropy:** entropy of Y remaining once X is observed, averaged over X . **Mutual information:** bits of shared information, or reduction in uncertainty about Y from knowing X . **KL divergence:** extra bits to encode p with a code optimized for q .

2. Which distribution maximizes entropy for n fixed outcomes? Why?

Uniform. Setting $\partial/\partial p_i$ of $-\sum_i p_i \log p_i - \lambda(\sum_i p_i - 1) = 0$ gives all p_i equal. Intuition: any concentration of probability commits to something, reducing average surprise.

3. KL asymmetry: why it matters when q is a model and p is data.

Forward KL $D_{\text{KL}}(p||q)$ blows up where data has mass but the model does not, so it is mass-covering. Reverse KL is mode-seeking. Cross-entropy classification uses forward KL, which is why models cannot ignore tail classes without paying a huge loss.

Derivation / Math

4. $H(X)$ for biased coin $p = 0.25$.

$$H = -0.25 \log_2 0.25 - 0.75 \log_2 0.75 = 0.25 \cdot 2 + 0.75 \cdot 0.415 \approx 0.811 \text{ bits.}$$

Less than 1, the fair coin entropy, because the outcome is more predictable on average.

5. **Show** $\text{Gain}(S, A) = I(\text{label}; A)$.

$$\text{Gain} = H(S) - H(S \mid A) = H(\text{label}) - H(\text{label} \mid A) = I(\text{label}; A).$$

That is literally one of the definitions of mutual information.

6. **Minimizing cross-entropy is equivalent to minimizing KL to data.**

$$H(p, q) = H(p) + D_{\text{KL}}(p \parallel q).$$

Since $H(p)$ is independent of q ,

$$\arg \min_q H(p, q) = \arg \min_q D_{\text{KL}}(p \parallel q).$$

So cross-entropy loss directly minimizes KL between the model and the empirical data distribution.

Analysis: Why Does It Matter?

7. **XOR features: zero individual MI, perfect joint prediction.**

With balanced XOR, $I(\text{label}; X_1) = 0$ and $I(\text{label}; X_2) = 0$, but $I(\text{label}; X_1, X_2) = 1$ bit. An individual-scoring filter (UL2) computes univariate MI and sees nothing, so both features look useless alone. Wrappers, which evaluate feature subsets via a downstream model, are what catch this.

8. **Why ICA minimizes mutual information.**

ICA's premise: observed signals are linear mixtures of independent sources. Independence is equivalent to $I(X; Y) = 0$. So minimizing pairwise MI among recovered components is the operational definition of making them independent, stronger than PCA's "make them orthogonal / uncorrelated."

Exam-Style

9. **Trace each quantity across SL, OL, UL.**

Entropy: SL2 tree splits, OL3 max-entropy principle, and UL2 information-gain feature filters.

MI: SL2 information gain, OL1 MIMIC's dependency tree, and UL3 ICA objective. **KL:** SL3 cross-entropy loss, OL4 MLE, and UL4 t-SNE objective.

10. **From memory.**

$$H(X) = - \sum_x p(x) \log p(x),$$

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = H(Y) - H(Y | X),$$

$$D_{\text{KL}}(p||q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

11. NFL theorem in one sentence.

Averaged over all possible target functions, no learning algorithm outperforms any other, so generalization always rests on the algorithm's inductive bias being matched to the actual problem class. Full treatment: ../02_optimization_uncertainty/OL3_information_theory.md.